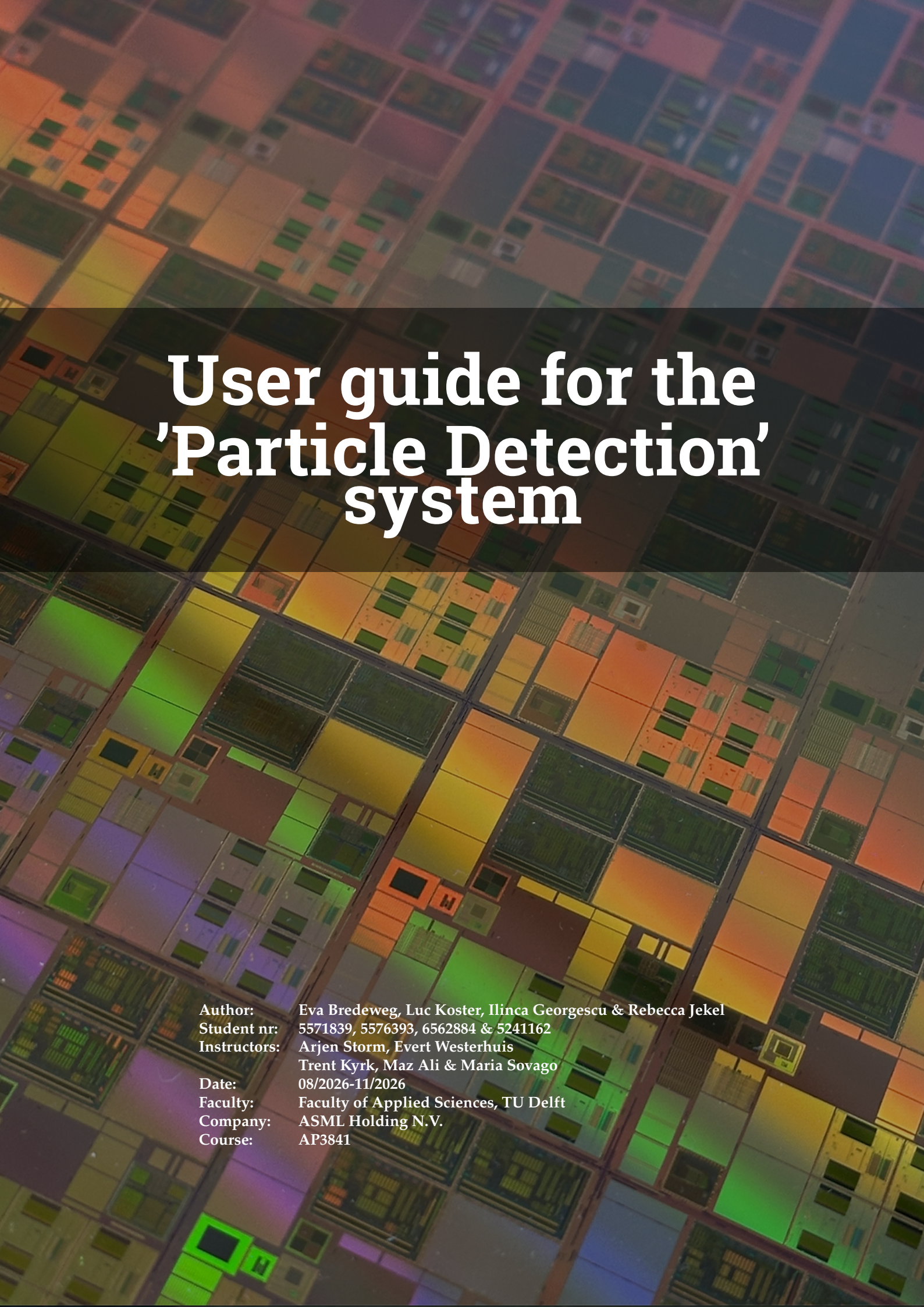


User guide for the 'Particle Detection' system

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1

Hardware

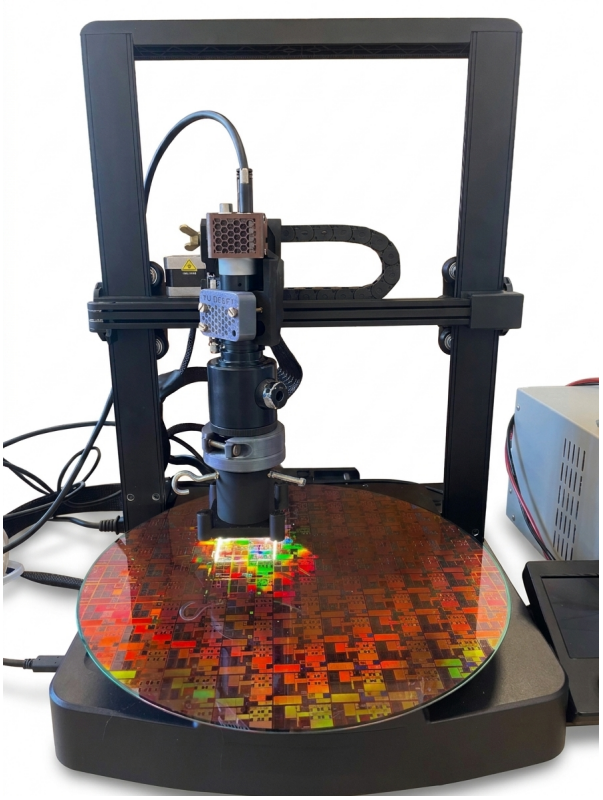


Figure 1.1: 3D printer setup. a) optical setup. b) moving gantry. c) mount. d) pcb+arduino. e) power supply. f) dark box.

1.1. Required Components

The main components of the system are described in detail in this section.

1.1.1. Optical Set-Up

Camera The Basler acA2040-90um camera [1] allows for high-resolution image acquisition. It connects via USB 3.0. Its resolution is 2048x2048px, it allows up to 90 frames/s, and it has a C-Lens mount.

Lens The Sill Optics S5LPL0015 lens [2] has a magnification of 1.53x, a focal length of 46.0mm and a working distance of 74mm.

Ringlight Ringlight V7 is used in the final prototype and results. Four 40mm COB LED strips [3] are configured in a square of 50x50mm centred around the FOV. A length-adjustable tube with an inner diameter of 30mm holds the LEDs 1-2mm above the wafer surface. The ringlight clamps to attach to the lens.

1.1.2. Movement Stage
A Creality Ender 3 V3-SE [4]is modified to be the movement stage. It is connected via USB-C to a PC.

1.1.3. Mount
Connection to moving gantry The mount connects the optical setup to the moving gantry. It is 3D printed in PLA and connects via 6 M6 screws. The connection is made at the neck of the lens via a custom-made clamp.

Rotational alignment tool A tool to align the optical set-up to the xy-movement stage.

1.1.4. Arduino & PCB
An Arduino UNO [5] connects to a circuit comprising four electronic switches for each COB LED strip.

1.1.5. Power supply
An external 12V DC power supply is needed with >500mA current capacity. None is supplied. In the system, a TENMA 72-10480 power supply [6] is used.

1.1.6. Computer
A computer on which the software is installed. The computer needs at least 3 USB connections.

1.1.7. Darkbox
A box that encompasses the movement stage and optical setup to prevent environmental light from reaching the camera sensor. It improves dark field imaging.

1.2. Safety & Handling
- Never touch the wafer surface. Only handle the wafer by its holder and non-critical areas.
- Set the power supply to 12V, <200mA before startup.
- Do not look directly into the lit ring light.
- Check for stuck or tight cables before starting a scan.
- Do not bump into the setup or the platform it is positioned on during a scan.
- Never lower the optical set-up without direct supervision. Prevent contact with the wafer surface.

1.3. Hardware Construction
The steps that need to be followed are given in order to prevent damage to the system. The usage of the software is explained in the next Chapter.

1. Set the power supply to 12V/200mA with its output switched off. Connect +12V to the red banana-plug input, and connect GND to the blue banana-plug input.
2. Connect the camera (USB 3.0), Arduino (USB 2.0), and movement stage (USB 2.0) to a computer.
3. Turn on the movement stage and power supply.
4. Insert the wafer onto the wafer holder.
5. Start up the software.

1.4. Alignment

The following steps finalise the physical set-up of the system.

1. Place the wafer in the holder on the movement stage using visual indicators. The middle 60x40mm of the wafer is scanned.
2. Bring the image into focus by moving the z-stage via the manual buttons in the software. A live view is provided to ease initial focusing. Adjust the length of the ring light if necessary using the three screws on the side to minimise the distance to the wafer. The lights should hover 1-2mm above the wafer surface.
3. If necessary, align the rotation of the optical set-up to the xy-stage. Image mm-paper in four tiles without overlap. Stitch them together and inspect visually for irregularities at seams. Turn the head screw of the mount accordingly and repeat this step until no irregularities are observed.

2

Software

The system software has two main functions: i) providing an understandable and flexible control interface for the hardware and ii) analysing the obtained pictures with the hardware.

The software is provided in two different formats, namely the 'Particle Detection' app and the 'Particle Detection Expert' app. The former app presents a simplified version of the software, containing only the 'Capture' and 'Analyse' modules, while the latter app presents the 'Verify' and 'Additional Tools' as well.

The normal system operation starts with 'Capture ', which outputs images, which can then be analysed with 'Analyse'. Verify can be used to verify the performance of the system.

2.0.1. Capture

'Capture' contains submodules controlling the lighting, the camera and the stage. Connection of all these hardware parts allows for the full 'Scan and capture' process to follow.

Setting up the lighting

Figure 2.1 presents a snapshot of the Lighting interface.

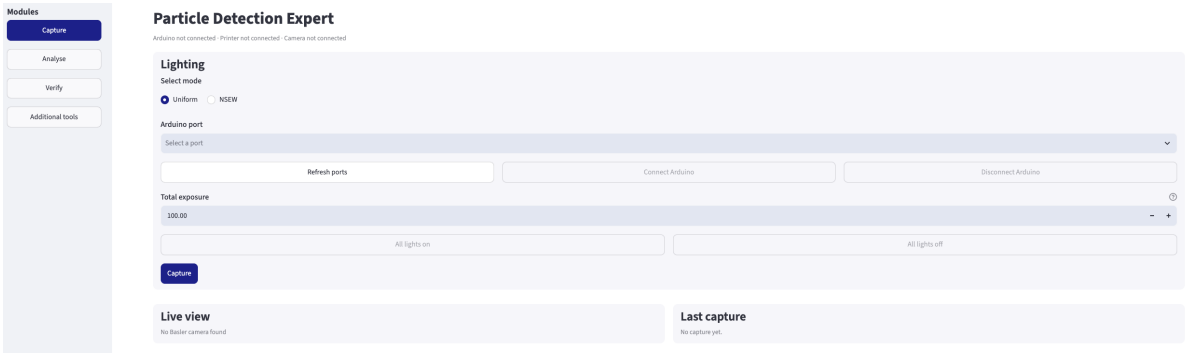


Figure 2.1: Setting up the lighting

1. Select the mode (Uniform or NSEW)
2. Select the port to connect the Arduino
3. Make sure the 'Live view' is running a live capture of your FOV
4. Make sure the interface prints 'Last reply: OK ALL'
5. Set the camera exposure time

Setting up the stage

Figure 2.2 presents a snapshot of the Stage interface. Once a stage is connected, the option for Manual Movement in x, y and z becomes available, with adjustable major/minor steps (see Figure 2.2). It can be used at any time outside of 'Scan and Capture' for incremental stage movement control.

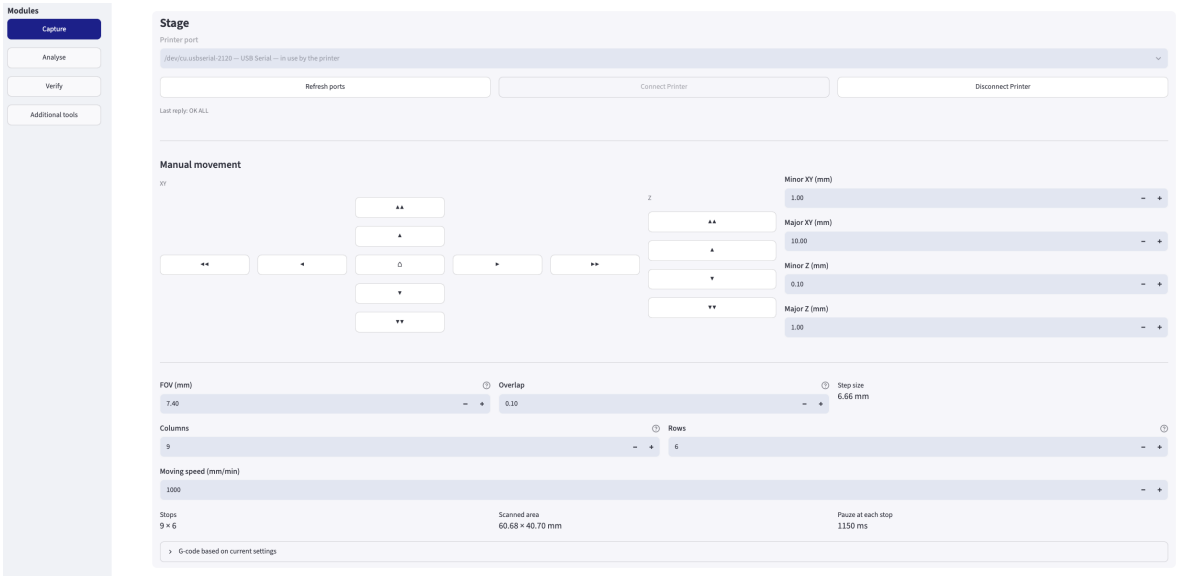


Figure 2.2: Setting up the stage

Steps for set-up:

1. Select the port to connect the stage
2. Make sure the interface prints 'Last reply: OK ALL'
3. Manually move the z-axis to bring your image in focus (watch the live view)
4. Check if the FOV, image overlap, and scanning pattern are correct

Autofocus

Figure 2.3 presents a snapshot of the Autofocus interface.

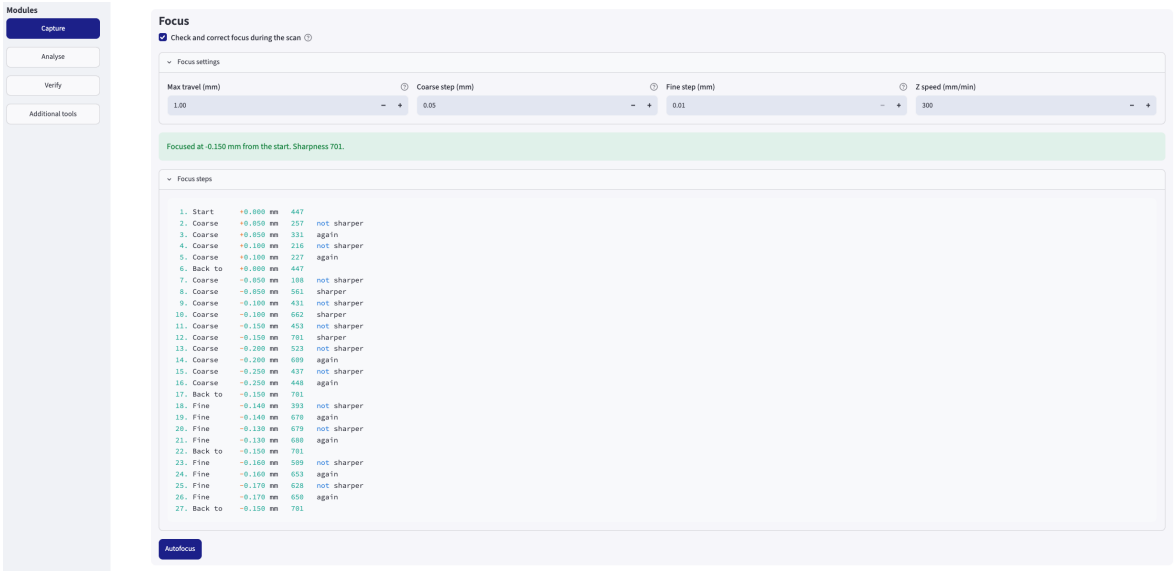


Figure 2.3: Working with the autofocus

1. Start the autofocus
2. Check in 'Live view' if the image is in focus
3. Before 'Scan and capture', select whether autofocus during the scan is preferred.

Scan and capture

1. Start 'Scan and capture'
2. Check if the files are saved into the 'Inputs' folder

2.0.2. Analyse

The standard settings for preprocessing and detection are different based on whether the uniform or NSEW approach is chosen. Therefore, first select the mode, then alter the rest of the parameters.

Loading parameters

Figure 2.4 presents a snapshot of the Loading interface.

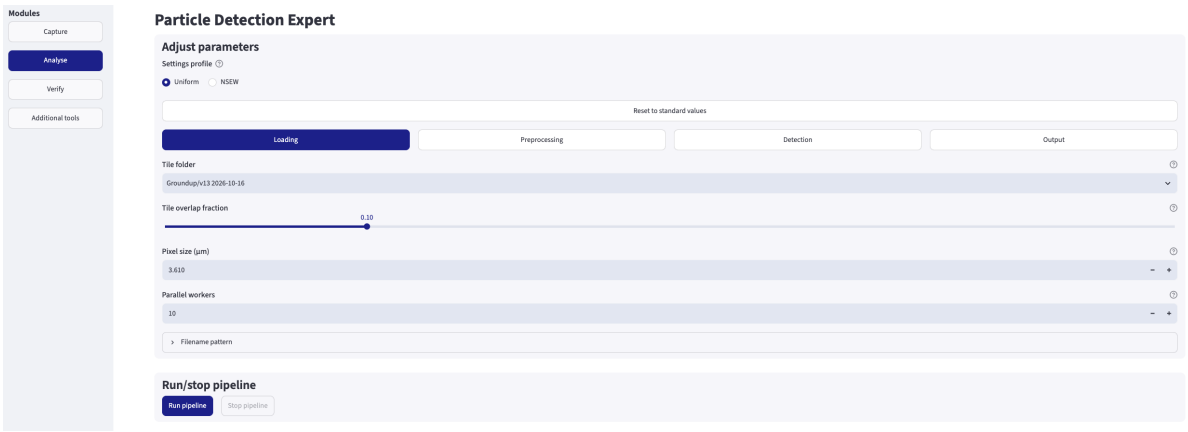


Figure 2.4: Setting up the parameters under 'Loading'

1. Select the path to the input folder
2. Select the overlap and pixel size
3. Select the parallel workers (CPU cores)
4. In case the pictures are not taken with 'Capture' and follow a different file name pattern, adjust accordingly

Note: the tile folder input gives suggestions when typing.
Note: the file name pattern is not shown directly, but can be altered as well. The standard setting is the pattern given when pictures are taken with 'Capture'

Preprocessing parameters

Figure 2.5 presents a snapshot of the Preprocessing interface.

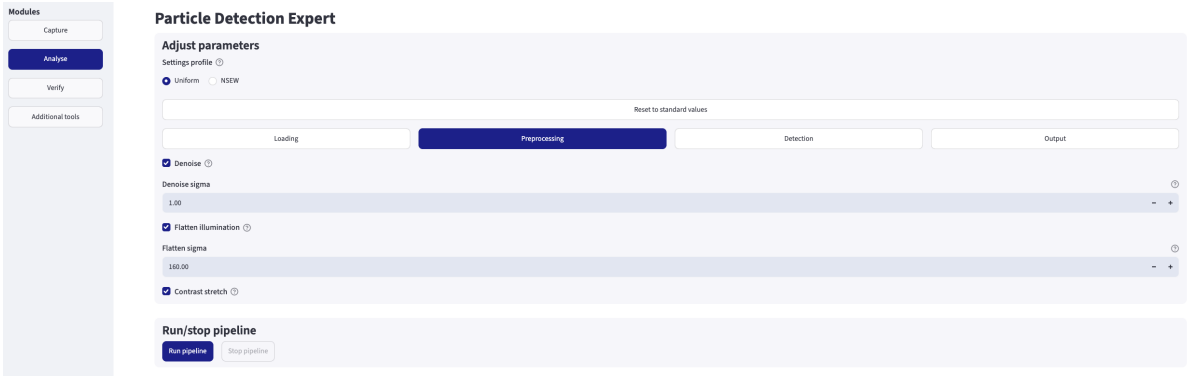


Figure 2.5: Setting up the parameters under 'Preprocessing'

1. Set the desired preprocessing parameters

Detection parameters

Figure 2.6 presents a snapshot of the Detection interface.

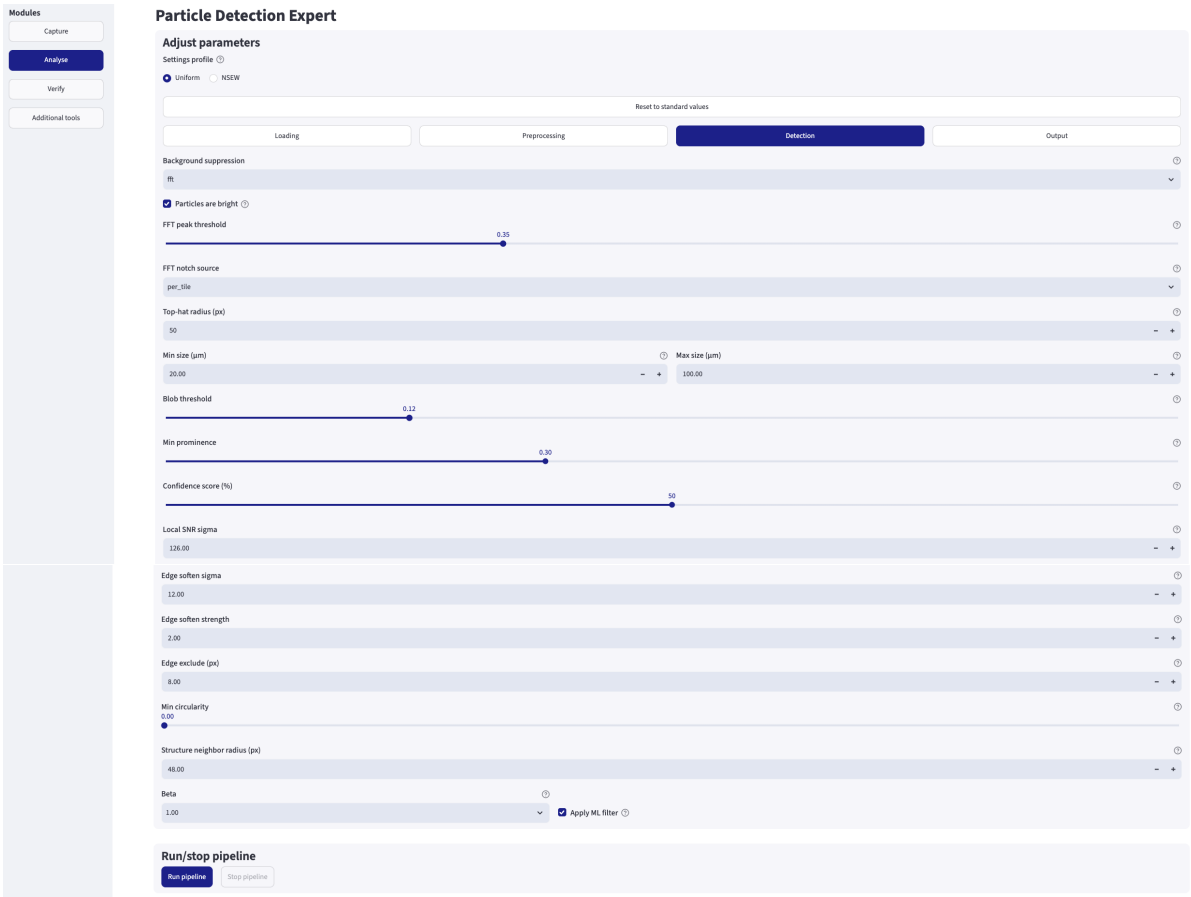


Figure 2.6: Setting up the parameters under 'Detection'

1. Set the desired detection parameters

Note: If brightfield images were to be taken instead of darkfield, the 'Particles are bright' mark can be unchecked. Note: the file name pattern is not shown directly, but can be altered as well. The standard setting is the pattern given when pictures are taken with 'Capture'

Output parameters

Figure 2.7 presents a snapshot of the Output interface.

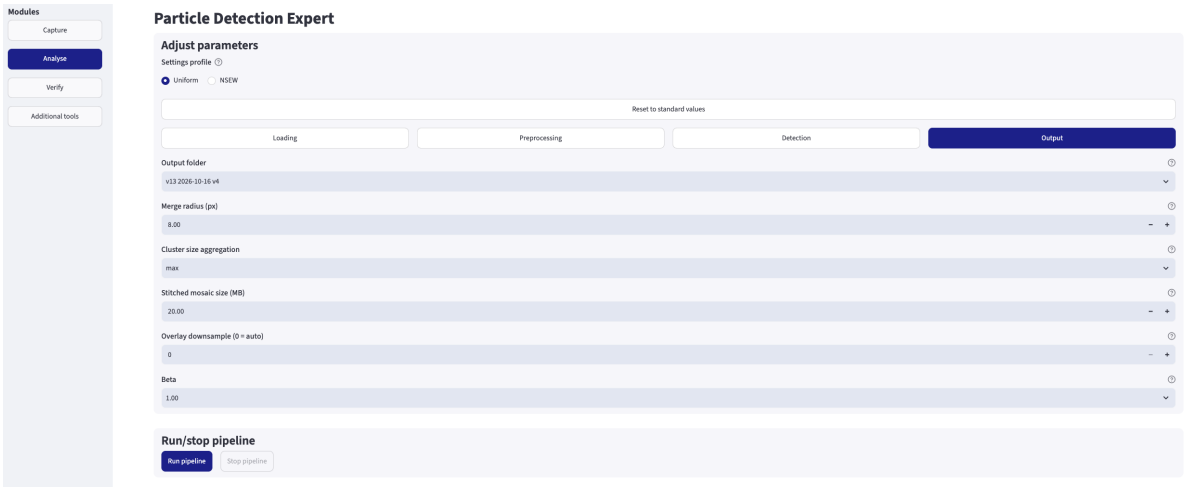


Figure 2.7: Setting up the parameters under 'Output'

1. Set the desired output parameters ('Output folder' is set automatically based on 'Input folder')

Given output

Summary in UI

When the pipeline is finished, the UI presents a summary of the last run, as shown in Figure 2.8.

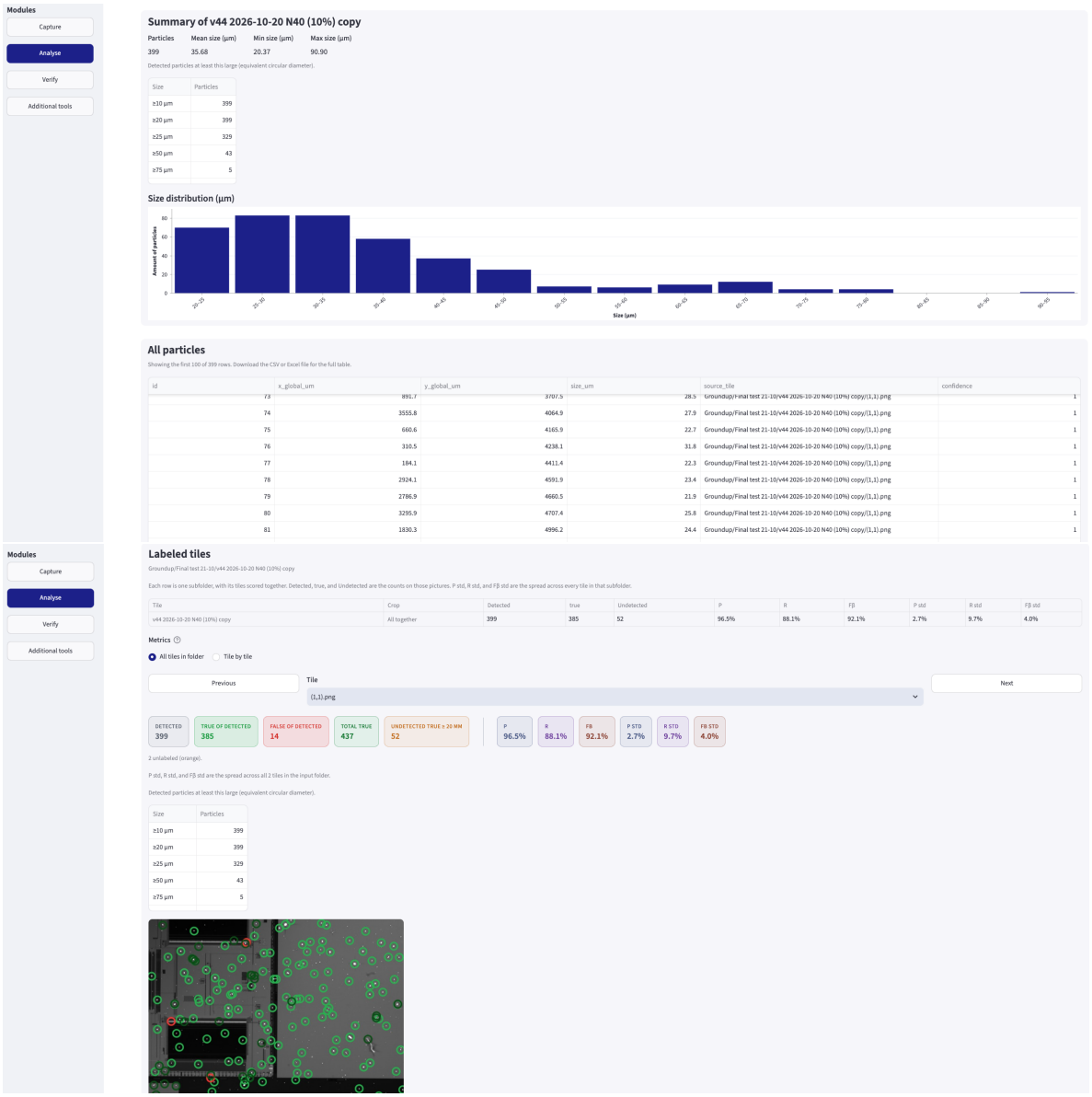


Figure 2.8: Snapshot of the output summary

The summary in the UI presents the following:

- Total number of detected particles
- Mean, min and max size of detected particles
- Number of particles ≥10 µm, ≥20 µm, ≥25 µm, ≥50 µm and ≥75 µm
- Histogram of size distribution
- First 100 hits with each hit's `id`, `x_global_um`, `y_global_um`, `size_um`, `source_tile` and `confidence`
- The stitched mosaic (standard with hit markings, can be unchecked)
- Labelled tiles with metrics (P, R, FB and their respective STDs) (relevant in case the hits have been labelled and undetected are marked)
- Crops of the hits (Figure 2.9)

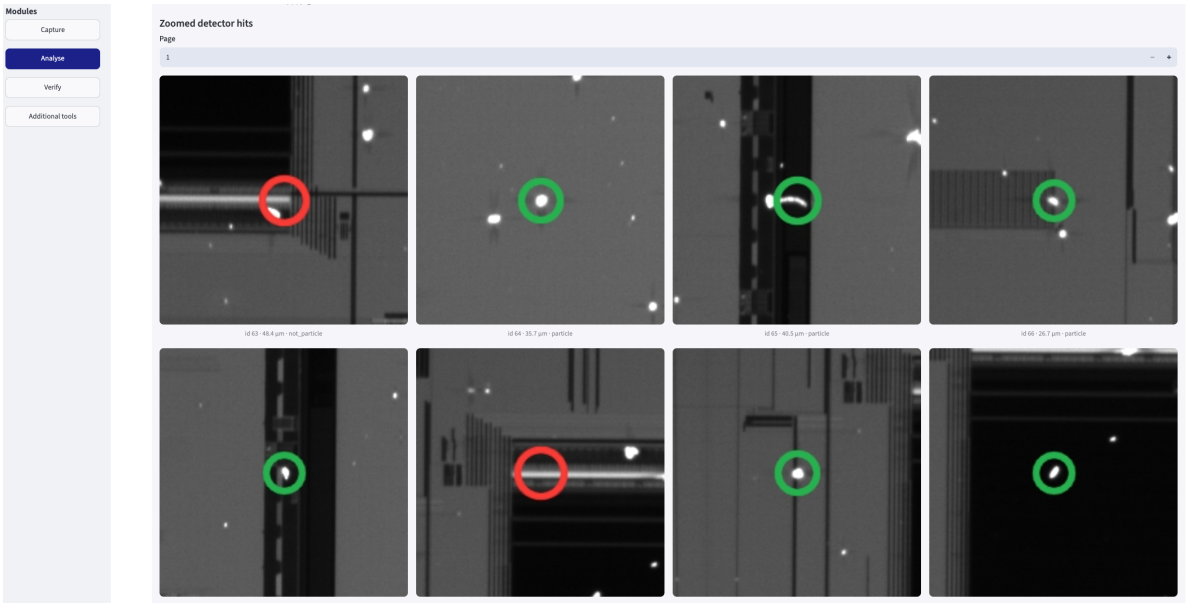


Figure 2.9: Snapshot of the hit crops

The metrics are only given when the hits have been labelled under 'Label' and missed particles are marked under 'Inspect' in the 'Verify' module.

Saved output files

Besides the summary in the UI, the software automatically saves files in the 'Output' folder it creates in the same place the software itself is loaded. These output folders are shown in Figure 2.10 and are:

- A JSON file with the set parameters in the last run (in the analysis)
- Two stitched mosaics, one with the hits labelled and one without the hits labelled
- Both a .csv and .xlsx file with each hit's `id`, `x_global_um`, `y_global_um`, `size_um`, `source_tile` and `confidence`
- A PDF with summary (Figure 2.11)

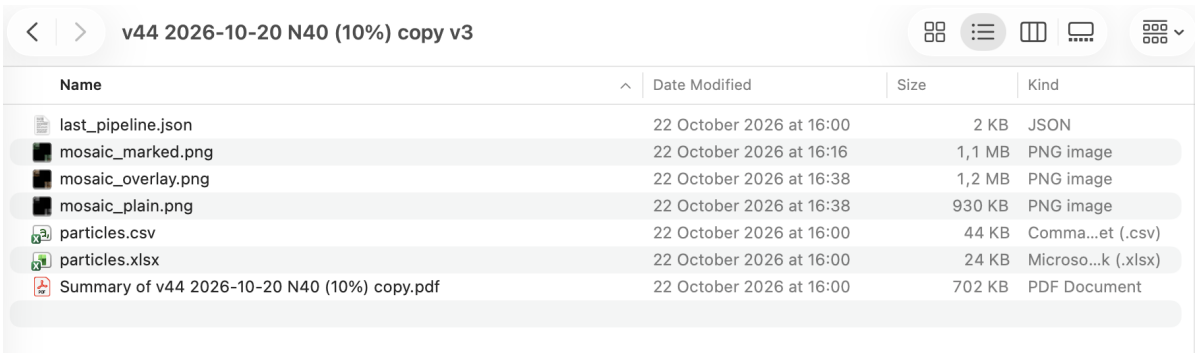


Figure 2.10: REMAKE Snapshot of all output files

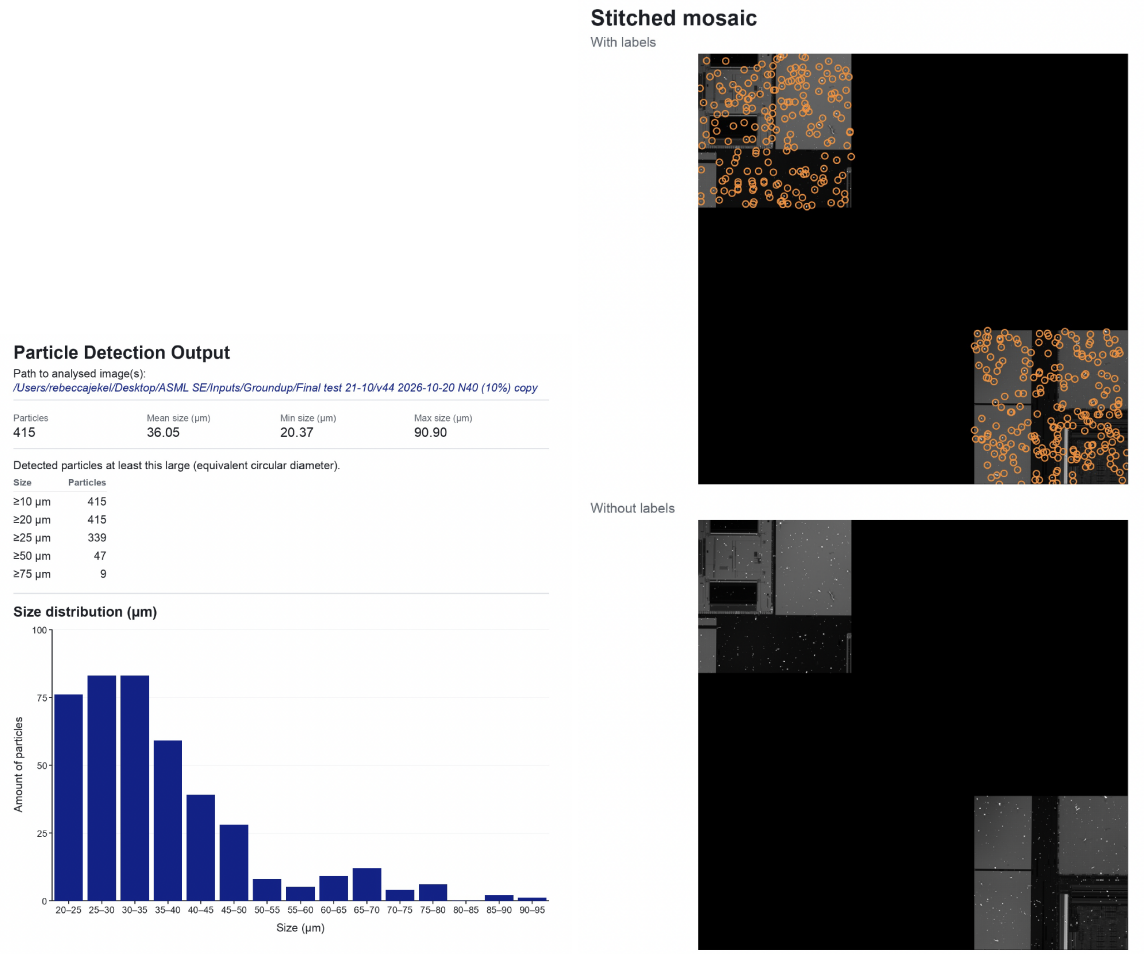


Figure 2.11: Snapshot of the output PDF

2.0.3. Verify

The goal of this module is to quantify how 'Analyse' is performing (calculate precision, recall and F_β. It consists of three submodules, Label, Inspect and Database.

Label

This submodule focuses on determining whether the hits are true particles or false particles in order to calculate the precision.

1. Click on 'Load last run'
2. Select labelling mode, namely 'One at a time' (Figure 2.12) or 'All hits' (Figure 2.13)
3. In case of 'All hits', select whether you are marking the particles (and saving the rest as not a particle) or vice versa
4. Label the hits. In case of 'One at a time', use the arrows on the keyboard and, in case of 'All hits' by marking the particles/not a particle
5. Continue until all hits are labelled
6. Click 'All done'

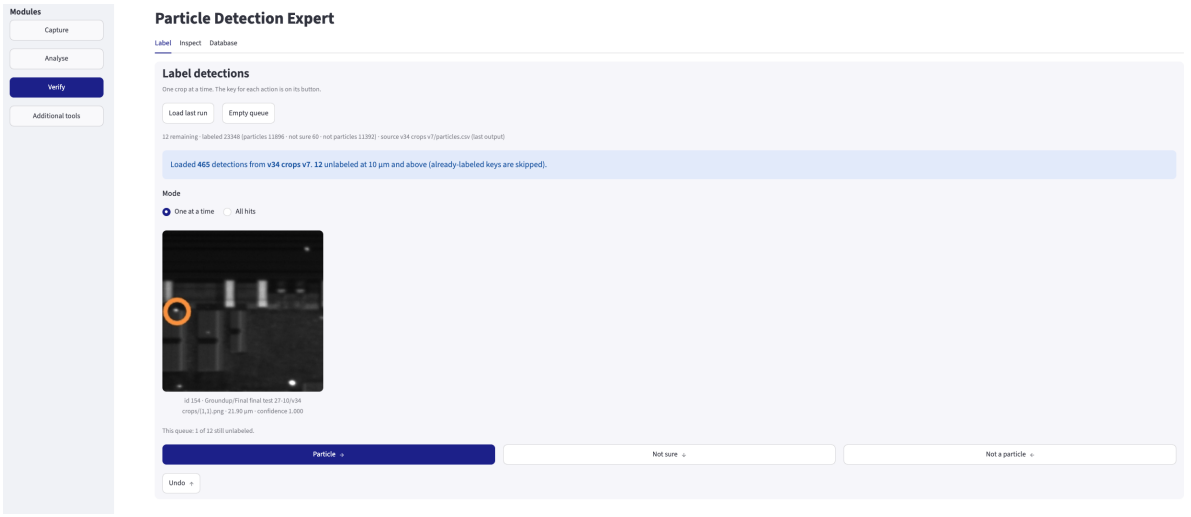


Figure 2.12: Labelling in 'One at a time' mode

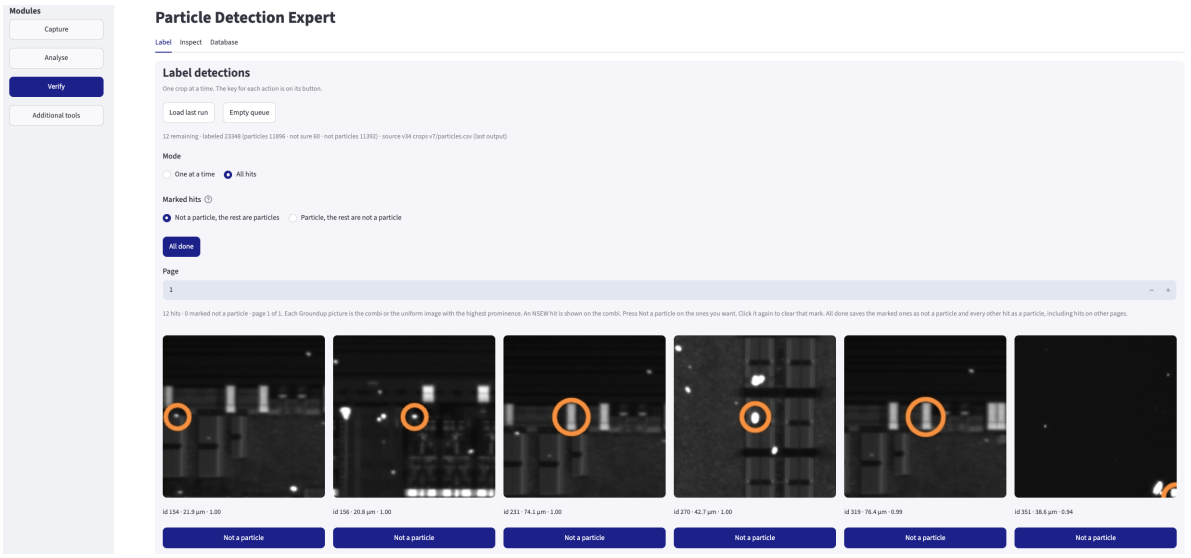


Figure 2.13: Labelling in 'All hits' mode

Note: Labelling in 'All hits' mode allows for quicker labelling if the majority of the hits are real particles or false particles.
Note: Clicking 'All done' marks the hits in the saved images in the output folder (from orange (unlabelled) to red (not a particle) or green (particle) and changes the precision in the summary under 'Analyse'.

Inspect

This submodule focuses on determining the number of undetected or missed particles in order to calculate the recall.

1. Select the right tile to inspect (with the dropdown or 'previous' or 'next')
2. Optionally select a crop (NW, N, NE, etc.) to inspect a zoomed-in tile
3. Scroll down to the 'Click to mark a miss' and click on the missed particle
4. Click on 'Mark missed particle'
5. Continue until all missed particles on all tiles are marked

6. Click 'All done'

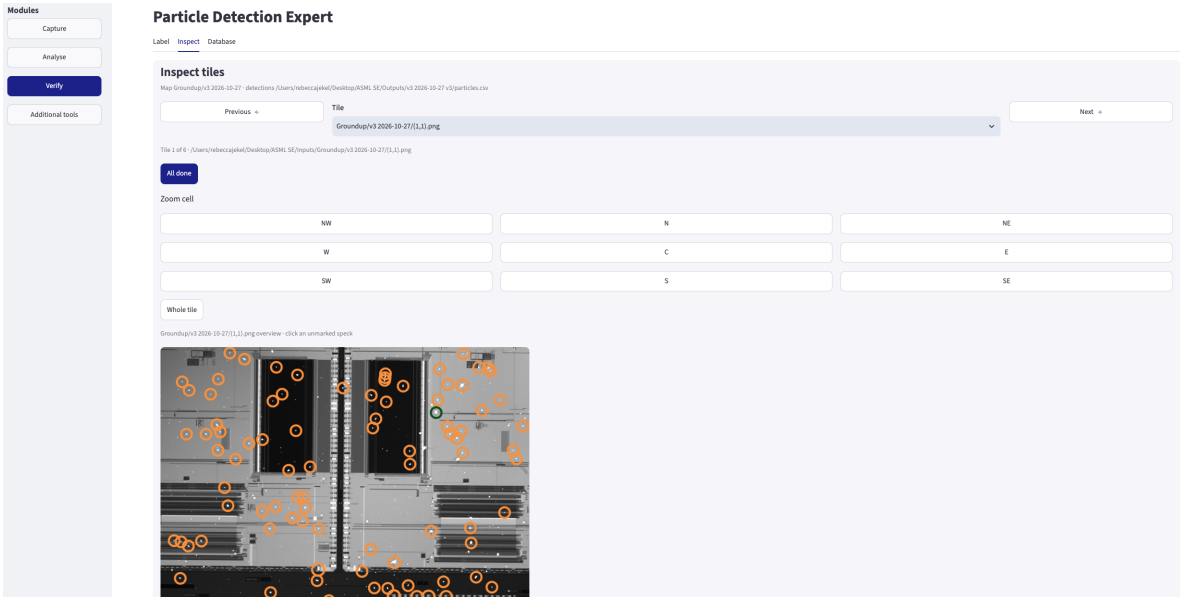


Figure 2.14: Overview of 'Inspect'

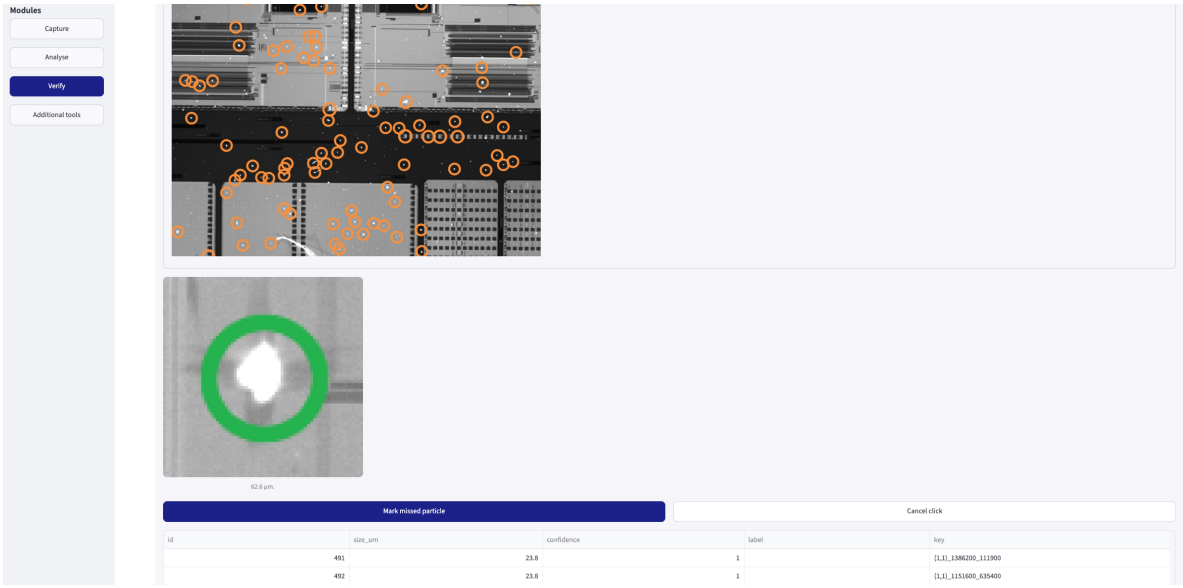


Figure 2.15: Adding missed marks in 'Inspect'

Note: Clicking 'All done' marks the missed particles in the saved images in the output folder (dark green circles) and changes the recall in the summary under Analyse.

2.0.4. Additional Tools

'Additional tools' consists of two submodules, 'Shadow' and 'Redirector'. Only the former is described here.

Shadow

The goal of this submodule was to extract the height from the particles, see Figure 2.16

1. Select whether the shadows will be measured from a particle that is in the database already or not

2. Select the desired parameters

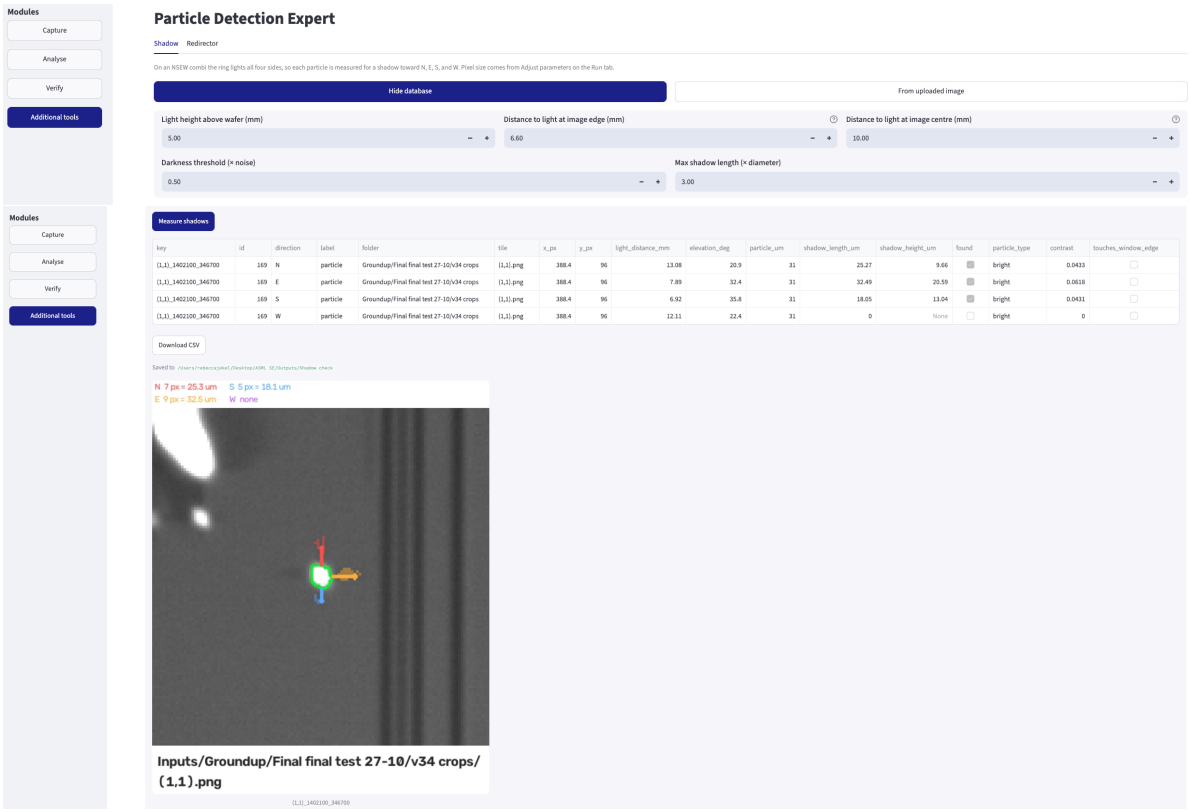


Figure 2.16: Overview of 'Shadow' results

Note: If 'From Database' is chosen, the key of a certain particle can be found under the crops or in the table under 'Database'.

3

Troubleshooting

Table 3.1: Common problems and solutions observed during operation of the system.

Problem	Potential Fixes
No live image	• Camera cable connected to a USB 3.0 port. • Restart the software.
Current limited	• Circuitry shorted. Check wiring.
Ring light off, or only some segments	• Power supply on and at 12V/200mA. • Check PCB connections and polarity. • Check Arduino USB connection.
Movement stage does not move	• Turn 3D printer on. • Check the right USB connection. • Check port selection in software.
Image goes out of focus during a scan	• The camera slowly sinks in z; use autofocus during the scan.
Holes between tiles in the stitched image	• Re-align camera rotation to xy-stage. • Check FOV and check if it corresponds in software.
Environment light enters	• Ring light is set too high. • Darkbox is missing/open.
Halos around particles	• Exposure time too long.
NSEW frames much darker than Uniform	• NSEW standard exposure is 800 ms; Uniform is 100 ms. • Wait for the gap between sides so the correct segment is lit.
Scan folder is not created	• The folder is created only after the camera accepts setup.
Tiles missing from the mosaic	• Filenames must be `(row,col).png`, or `(row,col)_N/E/S/W.png`. • Other names are still detected, but left out of the stitch.
Particles in the wrong place	• Set `overlap_fraction` to the overlap used in Capture. • Placement uses the filename grid, not image registration.
The same particle twice	• Raise `merge_radius_px`, or match `overlap_fraction` to Capture.
NSEW sides never combine	• A complete N/S/E/W set is required before the four frames become one tile.
Repeating pattern circled	• Raise `min_prominence` or `min_confidence`, or the model threshold.
Pad corners circled	• Keep the border drop on (8 px Uniform, 12 px NSEW).
Large flake missing	• Top-hat radius larger than the flake. • `max_size_nm` at least 100000.
Label queue empty	• Hits below 10 µm are left out. • Load last run on Verify.
Shadow height wrong	• Measure on an NSEW combi image. • Pixel size comes from the Analyse parameters.

References

[1] Basler AG. *Basler ace Classic acA2040-90um*. URL: https://www.baslerweb.com/en/shop/aca2040-90um/ (visited on 11/04/2026).

[2] Sill Optics GmbH. *Sill Optics: Precision Optics and Photonics*. URL: https://www.silloptics.de/en/ (visited on 11/04/2026).

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[4] Creality. *Ender-3 V3 SE*. URL: https://www.creality.com/products/creality-ender-3-v3-se (visited on 11/04/2026).

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[6] Multicomp Pro. *72-10480 Adjustable DC Power Supply, 30 V, 3 A*. URL: https://nl.farnell.com/multicomp-pro/72-10480/power-supply-1ch-30v-3a-adjustable/dp/2251946 (visited on 11/04/2026).